



Farm House

One number, many builds — variables

변수로 집과 농장을 함께

BIG PICTURE

Build in stages



House first, then walk over, then a farm patch beside it.

집을 짓고 → 옆으로 가서 → 농장

HOW TO COUNT

Count the axis — one block, one number



1 Stand on the start block — that spot is $(0, 0, 0)$.

2 $x \rightarrow$ walk **right**. Add **+1** for every block.

3 $y \rightarrow$ jump **up**. Add **+1** for every block.

4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is a variable ?

A variable is a **name that stores a number**. Change it once, and every loop that uses it changes too.

```
set size to 3  
set height to 2
```

size remembers **3**. **height** remembers **2**.

변수 = 숫자를 담는 이름표

CONCEPT

Use the variable in the loop

```
set size to 3

repeat 4 times:
  repeat size times:
    place block down
    move forward
  turn right
```

Change `size` to 5 and the whole house grows — no other edits.

size 하나만 바꾸면 전체가 커져요

CONCEPT

Stages keep builds tidy

1

Stage 1

house walls + roof

2

Stage 2

walk + farm patch

단계로 나누면 깔끔해요

CODE

Blocks: a variable-sized house

```
set size to 3
set height to 2

repeat height times:
  repeat 4 times:
    repeat size times:
      place block down
      move forward
    turn right
  move up by 1
```


CODE

Variables in Python

```
size = 3

for side in range(4):
    for step in range(size):
        agent.place(DOWN)
        agent.move(FORWARD, 1)
    agent.turn(RIGHT_TURN)
```

`range(size)` uses the variable. Change `size`, change everything.

변수 = variable

PREDICT 🤖

What does this build, and how tall?

Predict first — then click reveal

```
set size to 3
set height to 2
repeat height times:
  repeat 4 times:
    repeat size times:
      place block down
      move forward
    turn right
  move up by 1
```

Reveal output 🔍 reveal

VOCAB

New words today

variable a name that stores a value

set give a variable its value

stage one part of a multi-step build

patch a small square of farmland

변수

설정하다

단계

밭

What's wrong with this code?

Goal: house and farm should use the **same** size.

buggy

```
repeat 4 times:    # house = 3
  repeat 3 times:
    place block down
    move forward
  turn right
repeat 5 times:    # farm = 5
  repeat 5 times:
    place block down
    move forward
  turn right
```



DEBUG

The bug

Bug: the sizes are **hard-coded** as 3 and 5, so changing one doesn't change the other. Use one `size` variable in both loops.

fixed

```
set size to 3
# house uses size
repeat 4 times:
  repeat size times: ...
# farm uses size too
repeat 4 times:
  repeat size times: ...
```

YOUR TURN!

Build a farm house

- ✓ Open this week's world
- ✓ Build the **house walls + roof** (Stage 1)
- ✓ Walk to the spot beside it (Stage 2)
- ✓ Build a farm patch on the ground there
- ✓ Send a photo of house + farm together



QUIZ

QUESTION 1 OF 3

What is a **variable**?

A turn command.

A kind of roof.

A type of block.

A name that stores a value.



QUIZ

QUESTION 2 OF 3

If both builds use `size`, changing `size` to 5:

Does nothing.

Makes both builds bigger at once.

Changes only the first build.

Deletes the farm.



QUIZ

QUESTION 3 OF 3

Why split the build into **stages**?

It makes the code run slower.

There's no reason.

It keeps each part clear and the farm lands beside the house.

Stages are required by Minecraft.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Farm house done!

Next: the final project — a village.

다음 주: 마지막 프로젝트, 마을 만들기